GNPS identity-disjoint — 17 methods (consumed development)
Recall@1 ladder (rows preset by R@1; cells show within-metric rank)

	R@1	R@2	R@3	R@5	R@10	R@20	MRR	Mean rank	Median rank	Macro AUROC	Macro AP (hist. AUPRC)	Micro AUROC	Micro AP (hist. AUPRC)
full23_v132	88.4%	94.0%	95.9%	97.5%	98.9%	99.8%	92.5%	1.42	1.00	94.9%	92.5%	95.9%	84.4%
weighted_spectral_entropy	87.4%	93.0%	95.3%	97.0%	98.7%	99.7%	91.7%	1.49	1.00	94.3%	91.7%	94.2%	75.6%
entropy_raw_public	87.3%	93.4%	95.7%	97.5%	98.9%	99.8%	91.8%	1.43	1.00	94.6%	91.8%	94.2%	71.9%
denoising_search_public	87.1%	93.4%	95.5%	97.4%	98.8%	99.8%	91.7%	1.44	1.00	94.5%	91.7%	94.2%	72.9%
p2b_noise_v1_frozen	87.1%	93.2%	95.5%	97.4%	98.8%	99.8%	91.6%	1.45	1.00	94.4%	91.6%	94.3%	75.1%
p2b_sqrt_cosine	87.0%	93.0%	95.2%	97.2%	98.7%	99.7%	91.5%	1.47	1.00	94.1%	91.5%	93.9%	74.7%
p2b_official_frozen	87.0%	93.1%	95.5%	97.4%	98.8%	99.8%	91.5%	1.45	1.00	94.3%	91.5%	94.2%	74.5%
p2b_unweighted_entropy	87.0%	93.1%	95.3%	97.1%	98.7%	99.7%	91.5%	1.48	1.00	94.1%	91.5%	93.9%	74.8%
neutral_loss_sqrt_cosine	86.8%	93.0%	95.4%	97.3%	98.8%	99.8%	91.4%	1.46	1.00	94.3%	91.4%	94.1%	74.4%
noise_v1	86.6%	93.0%	95.1%	97.2%	98.7%	99.8%	91.3%	1.46	1.00	94.1%	91.3%	94.4%	75.2%
spec2vec_gnps_public	86.3%	92.5%	95.2%	97.0%	98.7%	99.7%	91.0%	1.50	1.00	93.8%	91.0%	93.7%	73.0%
spec2vec_2026_retrained	86.1%	92.5%	95.2%	97.1%	98.6%	99.7%	90.9%	1.50	1.00	93.7%	90.9%	93.7%	72.7%
official_dreams	85.4%	92.3%	95.0%	97.0%	98.8%	99.8%	90.5%	1.50	1.00	93.7%	90.5%	93.2%	68.1%
cosine_greedy	85.0%	91.9%	94.4%	96.7%	98.6%	99.7%	90.2%	1.54	1.00	93.4%	90.2%	93.3%	73.9%
modified_cosine	85.0%	91.9%	94.4%	96.7%	98.6%	99.7%	90.2%	1.54	1.00	93.4%	90.2%	93.3%	73.9%
ms2deepscore_2x_public	79.1%	88.5%	92.5%	95.7%	98.4%	99.6%	86.4%	1.71	1.00	90.7%	86.4%	90.4%	68.9%
specembedding_official	75.3%	85.9%	90.9%	95.0%	98.0%	99.6%	83.7%	1.86	1.00	89.1%	83.7%	85.6%	49.9%

Rank (1 = best)
17
16
15
14
13
12
11
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8
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3
2
1